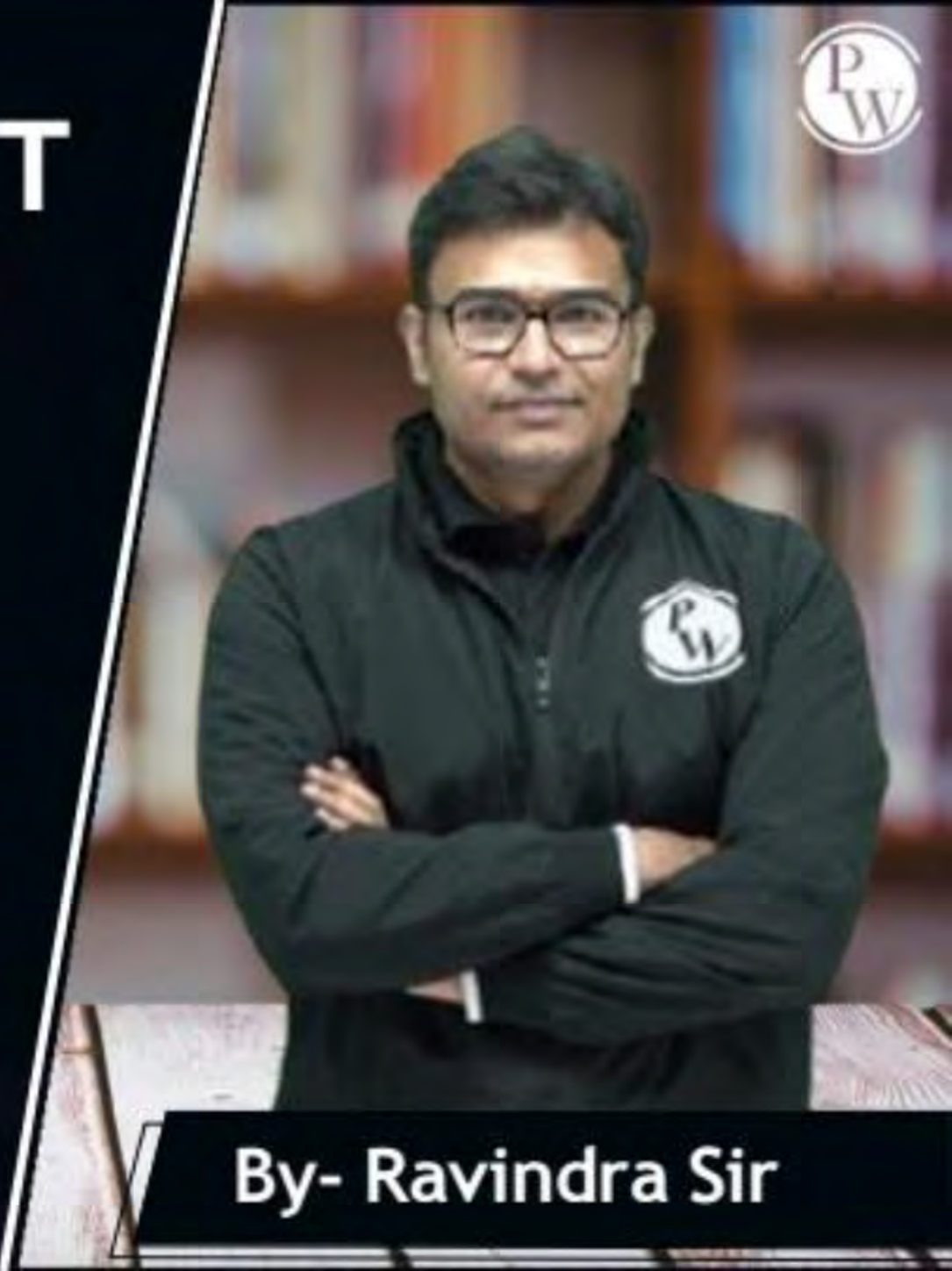
Computer Science & IT

COMPUTER NETWORK
(CN)

ISO-OSI Stack

Lecture No. 08



By- Ravindra Sir

Recap of Previous Lecture



Topic ARP

Topic

Topic

Topics to be Covered



- Topic
- Topic
- Topic

RARP
BOOTP
DHCP

Inspiring Stories : Sindhutai Sapkal



Background: Born poor in Maharashtra, faced abuse and homelessness.

Education: Studied later while doing odd jobs.

Achievements: Raised and educated over a thousand orphan children.

Impact: Known as “Mother of Orphans”; proved kindness changes lives.

Inspiring Stories : Rani Gaidinliu



Background: From a Naga village in Manipur.

Education: Village schooling.

Achievements: Led a tribal movement against British rule as a teenager.

Impact: Became a symbol of courage for Northeast India.

Inspiring Stories : Usha Mehta



Background: Born in Gujarat, loved the freedom movement.

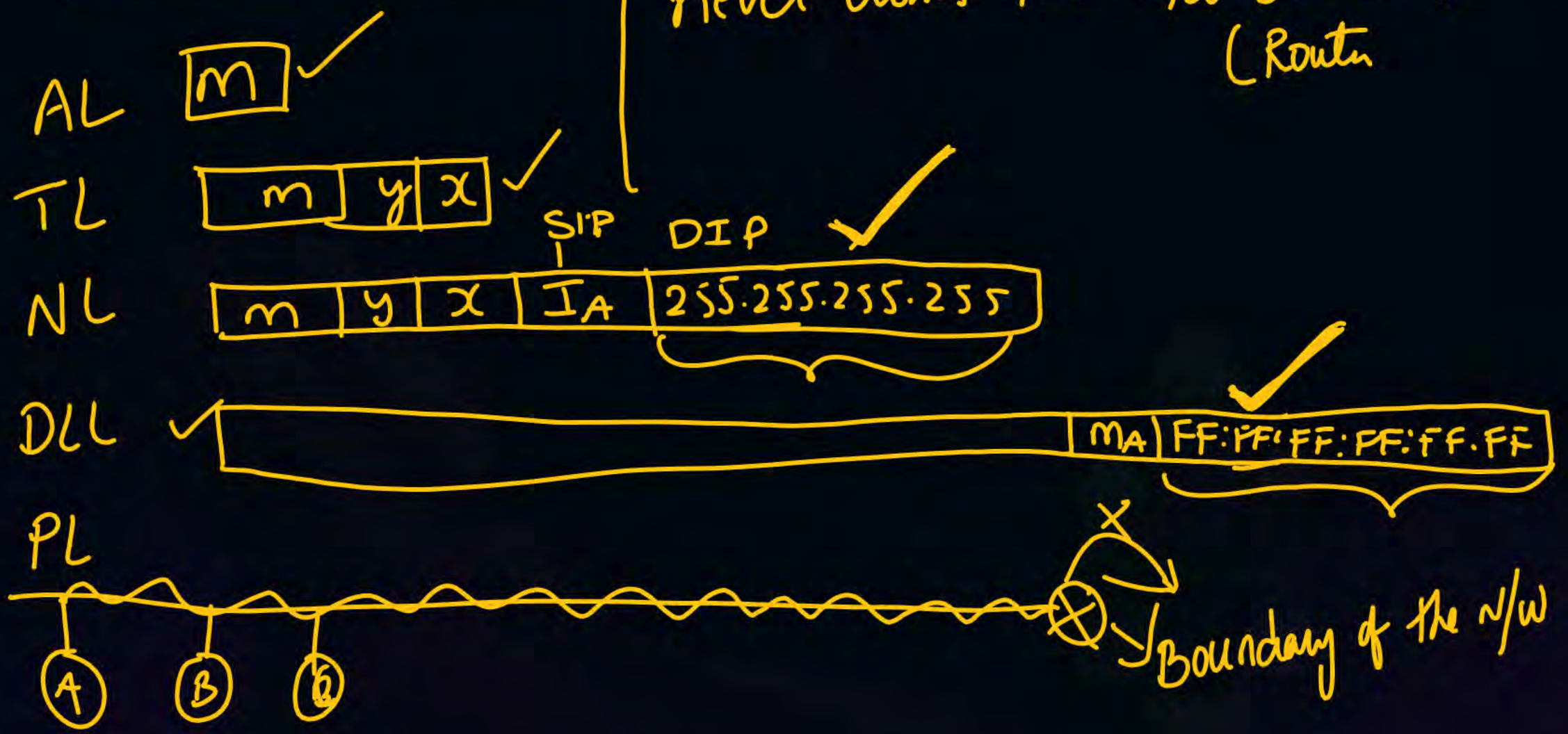
Education: Studied law.

Achievements: Ran a secret radio during Quit India (1942).

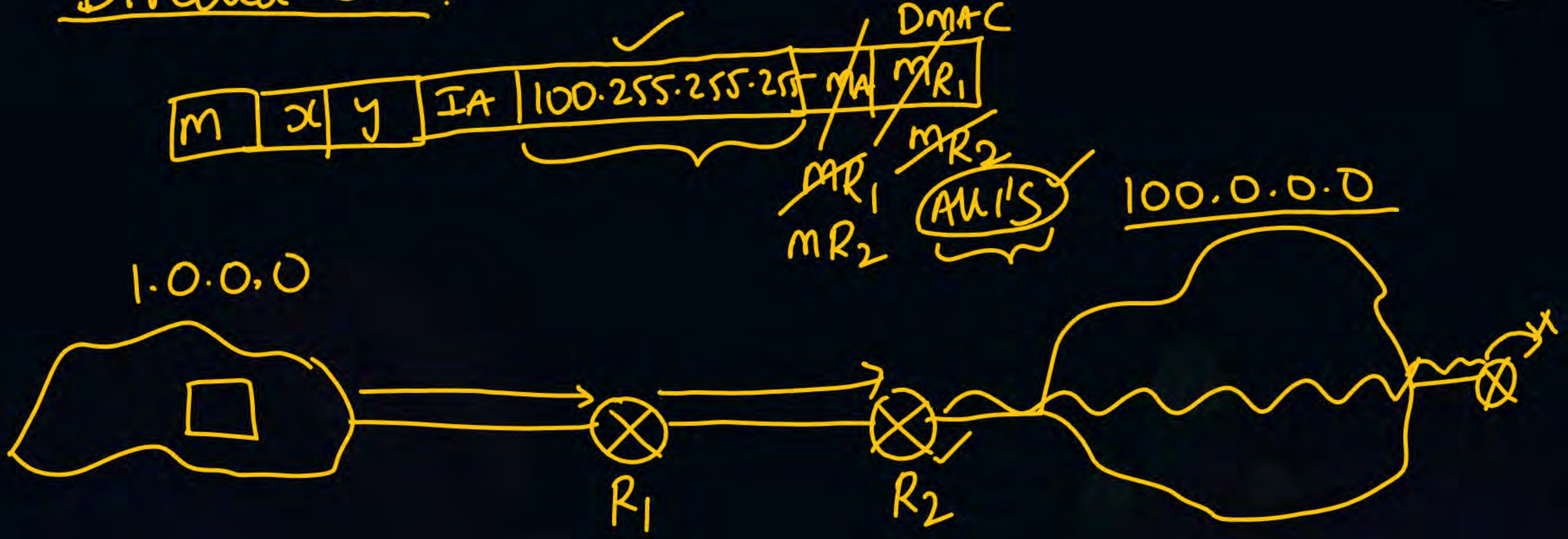
Impact: Kept the nation's morale high when leaders were jailed.

Limited Broadcasting :-

Broadcasting done at DLL, never crosses the N/w boundary (Router



Directed BC :-


m
x
y
IA
100.255.255.255
DMAC
MR1
MR2
DAR1
MR2
ALL'S
100.0.0.0
1.0.0.0
R1
R2

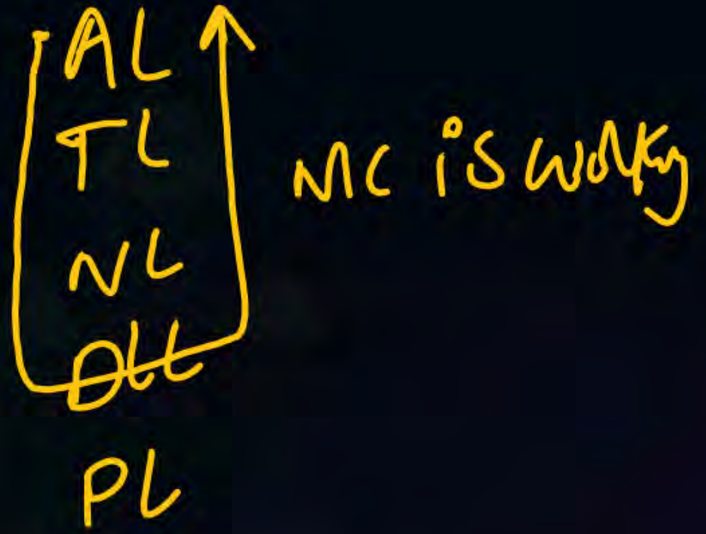
Special add 127:-



127.0.0.1 ✓

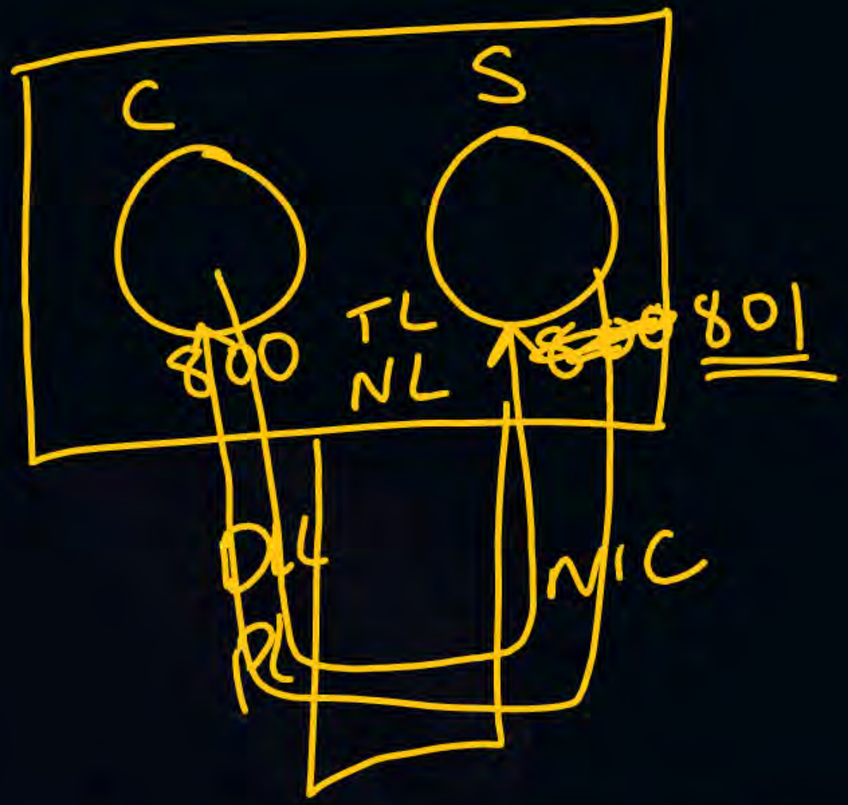
127.0.0.0 X

127.255.255.255 X



A packet with Destination IP 127.0.0.1, it will never go on to the N/W. It will go till NIC and Comes back

→ 127 is used to test self Connectivity.



Req	m	800	801	I_A	127.0.0.1

m	801	800	I_A	127.0.0.1

127 is used to test client Saver application in the Same host

Ping 127.0.0.1

RTT 5 ms → NIC is working

Time out → NIC is not working

RARP . Reverse ARP

ARP IP → MAC

RARP MAC → IP

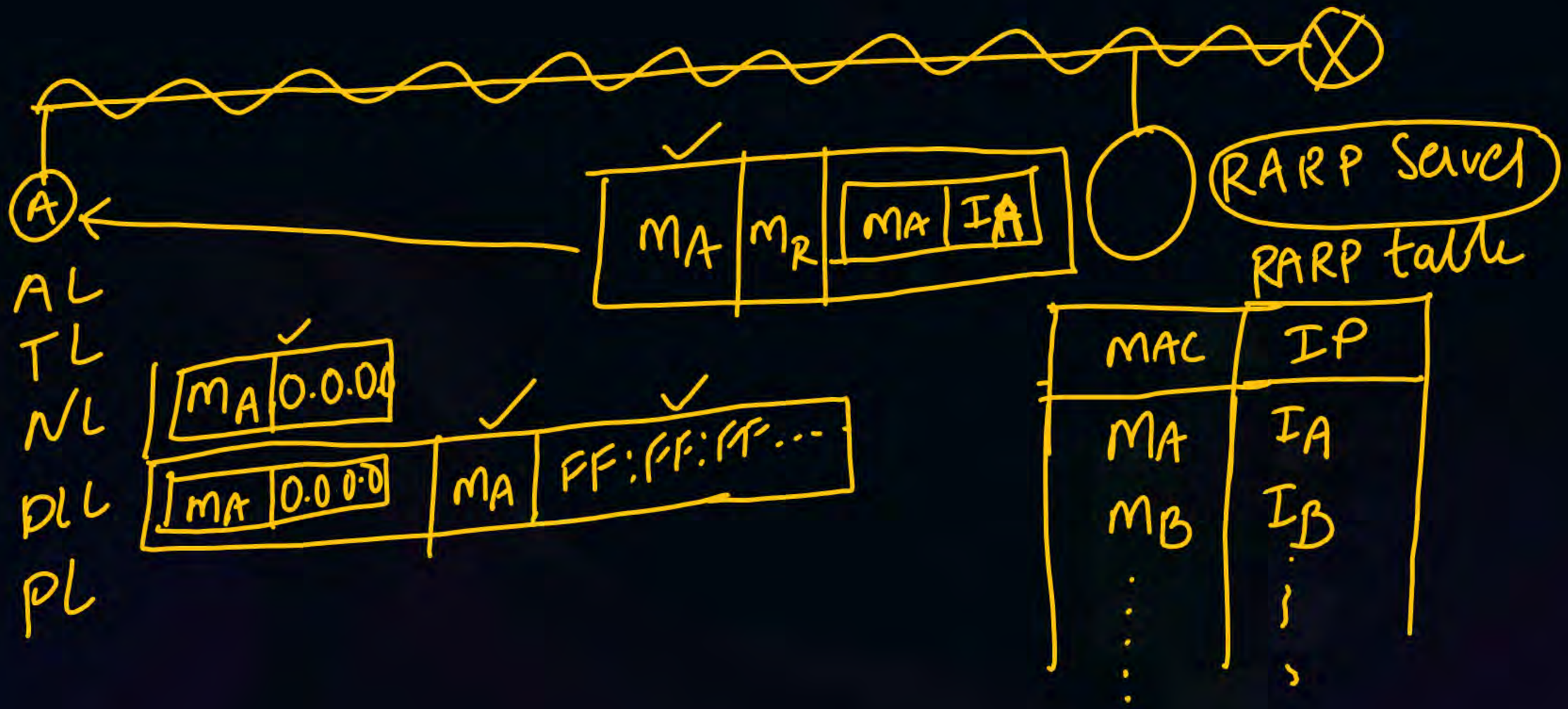
College N/W → NFS ✓

Host → no hard disk

IP ✗ ✓




NL
IP
RARP


A
AL
TL
NL
DLL
PL
MA | MR | MA | IA
MA | 0.0.0.0
MA | 0.0.0.0
MA
FF:FF:FF...
RARP Server
RARP table
MAC | IP
MA | IA
MB | IB

Even though we are calling RARP as a server, it is not actually a server. Because it runs at NL.

RARP is obsolete :-

Disadvantages of RARP :-

1) mapping table is static

MAC	IP
M_A	I_A
M_B	I_B
⋮	⋮

100 Hosts → 100 IP

At any time only 50 are online

2) RARP server is required in every n/w, because RARP Req is BC at DLL.



DB is distributed ✓

BOOTP:- (Runs at AL) (Provided by OS)

Bootstrap protocol



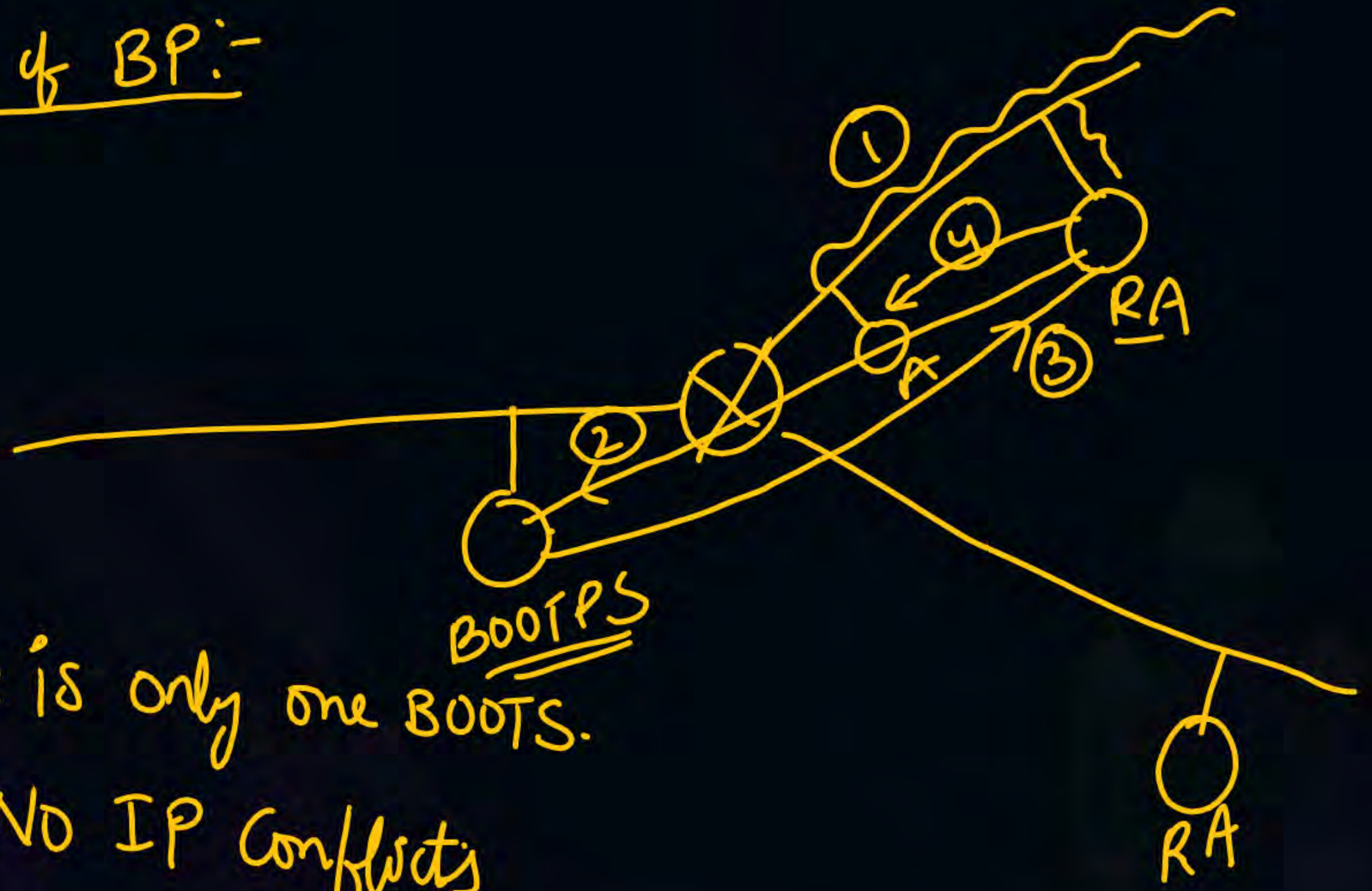
Disad:- mapping table is static

100 H → 50 H only at any

BOOT table

MAC	IP
MA	IA
MB	IB
⋮	⋮

Advantage of BP:-



There is only one BOOTS.

No IP Conflicts.

DHCP :- Dynamic Host Configuration protocol. (Runs at AL)

DHCP Server port : 67 ✓

DHCP client port : 68 ✓

100 H ✓

↓

50 H can online - 50 IP ✓



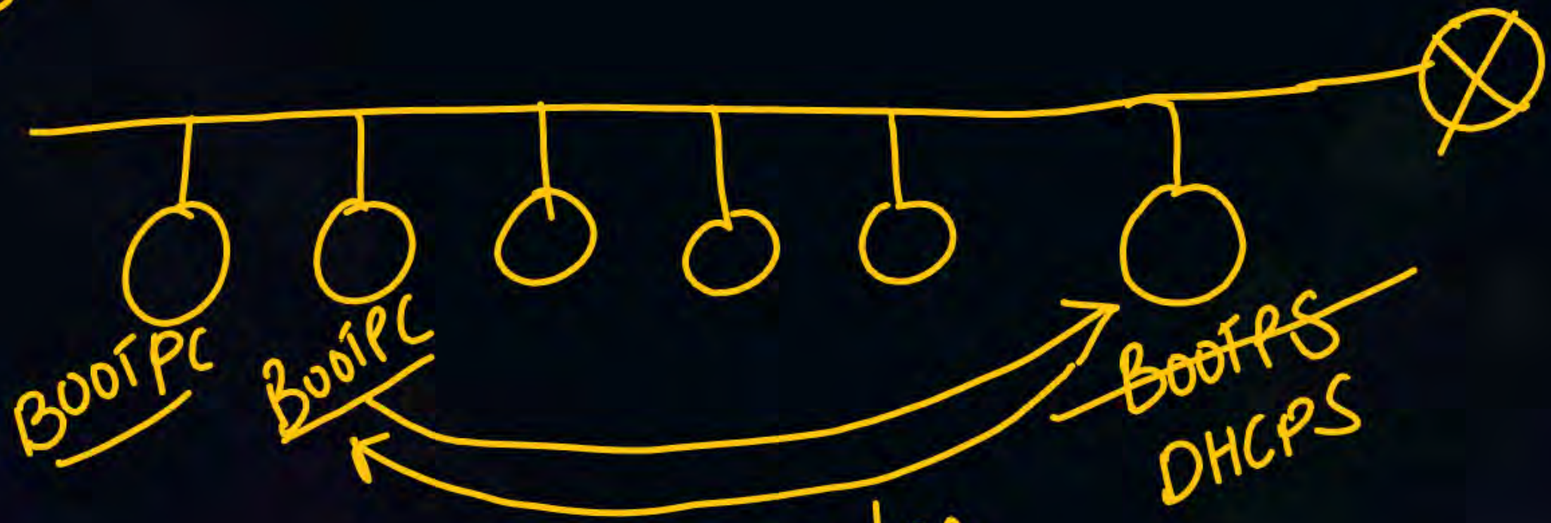
ADV :-



~~RARP~~ → BOOTP → DHCP

DHCP is back ward compatible with BOOTP:

ISP



BOOTP and DHCP has same port number

THANK - YOU